

The Chinese University of Hong Kong

ENGG1910: Demystifying Artificial Intelligence

Lab 2: Neural Translation Machine: Mandarin to Cantonese

Lab 2 Report

Answer the following questions and submit the report in a single PDF file onto Blackboard system. All relevant screenshot could be included in this report (no need to submit pictures as attachments on Blackboard).

1. Task 1

Obtain the encoded tokens of “然後我很快就睡著了。” (“After that, I soon fell asleep.”)

Answer:

A sample Chinese sentence:

然後我很快就睡著了。

Encode the sentence (a list of characters) into a list of input ids:

[101, 13792, 9331, 9970, 9327, 9425, 8485, 15348, 18616, 5028, 3566, 102, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]

Decode a list of input ids back to a list of characters:

[CLS] 然後我很快就睡著了。 [SEP] [PAD] [PAD] [PAD] [PAD]
[PAD] [PAD] [PAD] [PAD] [PAD] [PAD] [PAD] [PAD] [PAD]

2. Task 2

Evaluate the performance of the model before and after fine-tuning

- Translate the Mandarin sentence “她是一個非常美的女人。” using the two models and **report the BLEU scores of both translations.**

Answer:

Translation: 她是一個非常美的女人。

BLEU score: 0.1459819203086565

Translation: 佢係一個好靚嘅女人。

BLEU score: 1.0

- Use the same test set, “test_100.txt”, to evaluate and compare the pre-trained model and “final_model_300” by checking the average BLEU scores for each case. **Report the two BLEU scores for the pre-trained model and the model “final_model_300”.**

Answer:

Average BLEU score of 100 sentences: 0.16467238396311945

Average BLEU score of 100 sentences: 0.29806284506439346

3. Task 3

Evaluate the performance of the fine-tuned “final_model_300” on sentences with different lengths.

- Sentence Length (SL): number of characters of the sentence. E.g., the length of “然後我很快就睡著了。” (“After that, I soon fell asleep.”) is 10: “然”, “後”, “我”, “很”, “快”, “就”, “睡”, “著”, “了”, “。”.
- Refer to another two test sets, each containing 20 sentences.

sentences_8-10.txt: SL ranges from 8 to 10.

sentences_20-30.txt: SL ranges from 20 to 30

- **Report the two average BLEU scores on the two test sets** and see if Sentence Length (SL) affects the model’s performance.

Answer:

sentences_8-10.txt:

Average BLEU score of 20 sentences: 0.22280350320254935

sentences_20-30.txt:

Average BLEU score of 20 sentences: 0.36398515516368957

4. Task 4

Evaluate the performance of models trained with different amounts of data.

- Train another model with 900 sentences, “final_model_900”
- Obtain the output of the newly fine-tuned model using the same example test sentence, “她是一個非常美的女人。” (“She is a very beautiful woman.”), and **report the BLEU score**.

Answer:

Translation: 佢係一個好靚嘅女人。

BLEU score: 1.0

- Use the same test set (test_100.txt) to evaluate the model and **report the average BLEU score**.

Answer:

Average BLEU score of 100 sentences: 0.395502914376585

- **Compare the BLEU scores with the ones that you obtained in Task 2 using “final_model_300”**. Observe if the two models have different performance. If yes, **try to explain why**.

Answer:

Yes.

More Training Data: final_model_900 was trained on 900 sentence pairs, which is three times more data than final_model_300

Better Vocabulary and Grammar Mapping: Neural Machine Translation models rely heavily on data size. With more examples, the model learns a wider range of Cantonese vocabulary, slang, and grammatical differences from Mandarin.

Improved Generalization: Because final_model_900 has seen more sentence structures during training, it generalizes much better when translating new, unseen sentences in the 100-sentence test set, leading to higher translation accuracy and a higher BLEU score.

5. Task 5 (Optional)

Train an NMT model for *can2man*.

The NMT model in the demo is used to convert Mandarin to Cantonese. With the same model structure and parameters, train a model used to convert Cantonese to Mandarin and report the performance of the *can2man* model. **Report your basic idea and codes (in screenshot) that you might use for making this change.**

Original man2can: Reads index 0 (Mandarin) as the input features and index 1 (Cantonese) as the target labels.

Modified can2man: Swaps them so that index 1 (Cantonese) is the input features and index 0 (Mandarin) is the target labels.